view is just one color. The expansion essentially zoomed in on a part of the universe that was small enough to have already become the same temperature, and made the whole observable universe out of just that region.

Inflation also conveniently explains the density fluctuations, if we invoke a bit of quantum physics. The essential difference between the physics of the subatomic world and that of everyday life is that on the scale of individual particles, quantum mechanics imbues every interaction with an intrinsic, inescapable uncertainty. You may have heard of Heisenberg's Uncertainty Principle: it's the idea that there's a limit to the precision of any measurement, because the uncertainty built into quantum mechanics will always smear out the result in one way or another. If you measure a particle's position very precisely, you won't be able to determine its speed, and vice versa. Even if you just leave a particle alone, all of its properties will be subject to some amount of random shifting, and every time you measure it you could get a slightly different answer.

How does this tie into the cosmic microwave background? The hypothesis is that inflation was driven by a kind of energy field subject to quantum fluctuations: random jumps up and down. These fluctuations, which would normally only be ephemeral blips on a microscopic scale, change the density on the tiny scales where they happen, and then get stretched out to large enough regions to become substantial hills and valleys in the density distribution of the primordial gas. The little splotches we see in the cosmic microwave background make perfect sense if they are the natural evolution, over hundreds of thousands of years, of the fluctuations set down in the first 10^{-34} seconds of the cosmos. And those same little splotches are what eventually grew into all the galaxies and clusters of galaxies we see today.

The fact that the distribution of the largest structures in the universe can be exactly patterned on the minute wiggling of a quantum field is something that never fails to blow my mind. The links between cosmology and particle physics are never clearer, or more visually striking, than when we look at the cosmic microwave background.

But we're getting ahead of ourselves. The CMB is still, on these terms, eons away. We've only covered 10^{-34} seconds, and there's still plenty of story to tell.